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Network system and method of switching servers

Abstract

A device transmits operation information to a connected server. A second server transmits, to a first server, a request for first connection information for the device to connect to the first server based on reception of a registration request accompanying a firmware update from the device. The first server, after a reception of the request for the first connection information, starts transferring, to the second server, data related to the device, and transmits to the second server the first connection information in response to the request. The second server transmits to the device the first connection information received from the first server. When the transferring of the data related to the device reaches a predetermined step, second connection information for connecting to the second server is transmitted from either the first server or the second server to the device.

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Background/Summary

BACKGROUND
Technical Field
(1) One disclosed aspect of the embodiments relates to a network system to which a printer is connected via a network, the printer to which the connection is made is managed, and a job notification is performed by the network system, and to a method of switching servers.
Description of the Related Art
(2) The development and spread of the “Internet of Things” (hereinafter referred to as IoT) and cloud services have made it easier than before to collect data from various “things” via the Internet. Image forming apparatuses have become multifunctional, and multifunction peripherals called multifunction printers (hereinafter, referred to as MFPs) are utilized as client apparatuses.
(3) Recently, in view of protection of personal and confidential information, laws have been established to regulate the physical countries and geographical areas (hereafter referred to as regions) of client apparatuses that transmit data and servers that receive data. When attempting to connect to a server in a region different from regions of previous connections, a client apparatus takes measures, such as clearing connection settings, to prevent data from being leaked to the wrong region.
(4) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded

on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD®), or Blu-ray Disc®, a flash memory device, a memory card, and the like.

(5) The disclosure described in Japanese Patent Laid-Open No. 2013-168180 eliminates the need for the client apparatus to be aware of the change of a transmission destination by a pre-change server and a post-change server exchanging data of the client apparatus when the servers are switched. After the servers have performed the above-mentioned processing, the client apparatus is notified of the change of the server transmission destination for the first time by receiving connection settings from the post-change server.

(6) However, in the disclosure described in Japanese Patent Laid-Open No. 2013-168180, if the client apparatus connects to the pre-change server before connection settings for a connection destination change are received, an inconsistency will occur in client information transferred to the post-change server, resulting in cases where operation is not performed normally.

(7) As a method of solving such a problem, a method of preventing inconsistencies by blocking connections from the client device during data transfer is conceivable. However, when, for example, the client apparatus is a printer, cloud services (e.g., a cloud printing service) provided via the server will cease to be available during data transfer.

SUMMARY

(8) The disclosure allows client-related data transfer between servers, regardless of a state of a client apparatus and while services are continuously provided to the client apparatus during data transfer. In the following, the term "unit" may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term "unit" refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or program that, when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term "unit" refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. It may include mechanical, optical, or electrical components, or any combination of them. It may include active (e.g., transistors) or passive (e.g., capacitor) components. It may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. It may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term "unit" or "circuit" refers to any combination of the software and hardware contexts as described above. In addition, the term "element," "assembly," "component," or "device" may also refer to "circuit" with or without integration with packaging

materials. Furthermore, depending on the context, the term “portion,” “part,” “device,” “switch,” or similar terms may refer to a circuit or a group of circuits. The circuit or group of circuits may include electronic, mechanical, or optical elements such as capacitors, diodes, transistors. For example, a switch is a circuit that turns on and turns off a connection. It can be implemented by a transistor circuit or similar electronic devices.

(9) In order to solve the above-described problems, the disclosure includes the following configurations. According to one aspect of the disclosure, a network system includes a device, a first server, and a second server. The device transmits operation information to a connected server. The second server includes at least one second processor. At least one second program executed by the at least one second processor causes the second server to transmit, to the first server, a request for first connection information for the device to connect to the first server based on reception of a registration request accompanying a firmware update from the device. The first server includes at least one first processor. At least one first program executed by the at least one first processor causes the first server to, after a reception of the request for the first connection information, start transferring, to the second server, data related to the device, and transmit to the second server the first connection information in response to the request for the first connection information. The at least one second program causes the second server to further transmit to the device the first connection information received from the first server. When the transferring of the data related to the device reaches a predetermined step, second connection information for connecting to the second server is transmitted from either the first server or the second server to the device.

(10) By virtue of the disclosure, it is possible to perform client-related data transfer between servers, regardless of the state of a client apparatus and while services are continuously provided to the client apparatus during data transfer.

(11) Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an example of a configuration of a client-server type network system according to a first embodiment.

(2) FIG. **2** is a diagram illustrating a configuration of an application server **100** and management servers **200** and **300**.

(3) FIG. **3** is a diagram illustrating a configuration of a printer **400**.

(4) FIG. **4** is a sequence diagram illustrating processing for registering the printer **400** not having a server switching function in the management server **200**.

(5) FIG. **5** is a sequence diagram illustrating processing for registering the printer **400** having the server switching function in a management server **300**.

(6) FIG. **6** is a sequence diagram in which the printer **400** registered in the management server **300** re-obtains connection destination information.

(7) FIG. **7** is a sequence diagram illustrating an operation for when the printer **400** is connected to a management server for the first time in a case where it has never been connected to a management server and where the printer **400** has obtained the server switching function.

(8) FIG. **8** is a sequence diagram illustrating processing in which the printer **400**, after obtaining the server switching function, connects to a management server and transfers, to the management server **300**, printer information present in the management server **200** to which the printer **400** was connected before a ROM update.

(9) FIG. **9** is a flowchart of processing of the management server **300** after it has received a registration request.

(10) FIG. **10** is a flowchart of processing of the management server **300** after it has received a registration request.

DESCRIPTION OF THE EMBODIMENTS

(11) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the disclosure. Multiple features are described in the embodiments, but limitation is not made to an embodiment that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

(12) In the present embodiment, a cloud printing service (or a cloud printing system) will be described. In the present cloud printing service, a printer, an application server, and a management server work together to notify the printer of job data, which has been generated by the management server upon receipt of a print request from a web application running on the application server, and perform printing. A description will be given for an example in which in the cloud printing service, a printer that does not have a server switching function obtains the server switching function by a firmware update and switches over the management server to which it connects, triggered by the firmware update. The firmware update may also be called a Read Only Memory (ROM) update. The printer belonging to the cloud printing system may also be referred to, more generally, as a device.

(13) A management server that manages the printer belonging to the cloud printing service is one of the management servers that are present in each country or geographical area and is predetermined. The management server is determined by a shipping destination of the printer, information on the country or geographical area in which the printer is installed, a service usage status of a user, and the like to prevent printer-specific information and personal information of the user from being leaked to foreign countries and to optimize communication time. The server switching function is a function for, when a management server determined in advance for the printer is changed to another management server, transferring the printer registered in a pre-change management server to a post-change management server. The change of the management server here does not include a device update and the like and includes a change from a management server in which the printer has been registered to another management server. That is, a server switch refers to a change of a destination server that the printer accesses at the time of cloud printing.

(14) System Configuration

(15) FIG. **1** is a diagram illustrating an overall configuration of a data transmission system according to the present embodiment. First, a configuration of a system in which an application server **100**, management servers **200** and **300**, a printer **400**, and a global database **600** are linked will be described. A network **1000** is a network that connects the application server **100**, the management servers **200** and **300**, and the printer **400** to each other. The network **1000** may be, for example, the Internet, and is a communication network realized by a local area network (hereinafter, abbreviated as LAN), a wide area network (hereinafter, abbreviated as WAN), or the like. The network **1000** may further include wired communications, such as a telephone line, a dedicated digital line, an ATM and frame relay line, and a cable TV line, and wireless communications, such as a wireless line for data broadcasting. The network **1000** may be of any type so long as it enables exchange of data between the respective elements. In the present embodiment, a description will be given assuming that the network **1000** is the Internet.

(16) A network **1100** is a network that connects the management servers **200** and **300** and the global database **600** to each other. The network **1100** may be, for example, the Internet, and is a communication network realized by a LAN, a WAN, a telephone line, a dedicated digital line, an ATM and frame relay line, a cable TV line, a wireless line for data broadcasting, and the like. The network **1100** may be of any type so long as it enables exchange of data between the respective

elements. In the present embodiment, a description will be given assuming that the network **1100** is the Internet.
(17) A database **110** of the application server **100** stores various kinds of information for providing the cloud printing service to the user. A web application **120** transmits a print request to the management server **200** or **300** via the network **1000** and receives operation information of the printer **400** from the management server **200** or **300**.
(18) Databases **210** and **310** of the management servers **200** and **300** store client information for the printer **400** to establish a connection and print job information of the printer **400**. Control applications **220** and **320** generate job data based on a print request received from the application server **100** and transmit the job data to the printer **400** via the network **1000**. The control applications **220** and **320** also transmit a job notification to the printer **400** via the network **1000**. In the present embodiment, the management server **200** and the management server **300** belong to regions that are different from each other and, for example, manage printers that are in their respective regions and provide job data, which is data to be printed, to the printer.
(19) A communication module **410** of the printer **400** receives a job notification by communicating with a management server in which the printer **400** is registered between the management server **200** and the management server **300** via the network **1000**. Furthermore, the printer **400** receives job data from the management server **200** or **300** via the network **1000** using job data connection information included in the job notification. A storage device **430** stores job data. A print module **420** prints job data received and stored by the communication module **410** by processing the job data.
(20) The global database **600** is connected to the management server **200** and the management server **300** via the network **1100** and manages registration of information shared between the management servers. In FIG. **1**, each element is illustrated in a single-unit configuration for the sake of descriptive simplicity in the present embodiment; however, there is no intention to limit the number of components of the elements in the configuration. Each element may be constituted by one or more components.
(21) The management server **200** may be referred to as a first server, and the management server **300** may be referred to as a second server. The data transmission system of FIG. **1** may be referred to as a network system or a cloud printing system.
(22) Hardware Configuration of Each Apparatus
(23) FIG. **2** is a block diagram illustrating a schematic configuration of the application server **100** and the management servers **200** and **300**. A Central Processing Unit (CPU) **301** is a circuit or device for controlling each of the following units. A disk unit or circuit **302** stores an application program **312**, a database **313**, an Operating System (OS), and various other files to be read out by the CPU **301**. An external storage medium reading unit or circuit **303** is an apparatus for reading out information, such as a file stored in an external storage medium such as a Secure Digital (SD) card. A memory **304** is configured by a Random Access Memory (RAM) or the like, and the CPU **301** temporarily stores and buffers data and the like as necessary. A display unit or circuit **305** is configured by, for example, a Liquid Crystal Display (LCD), and displays various kinds of information. An operation unit or circuit **306** includes a keyboard, a mouse, and the like for the user to perform various kinds of input operations. A network communication unit or circuit **307** is connected to a network, such as the Internet, and performs various kinds of communication. The communication unit or circuit **307** supports a wired LAN and a wireless LAN. Each of the above-described units or circuits is connected to each other by a bus **309**. The web application **120** of the web application server **100**, the control application **220** of the management server **200**, and the control application **320** of the management server **300**, which will be described later, are realized by the CPU **301** reading and executing a program necessary for processing.
(24) FIG. **3** is a diagram illustrating a configuration of the printer **400**. A CPU **401** performs computation, determination, and control of data/instructions according to a program stored in a

RAM **402** or a ROM **403**. The RAM **402** is used as temporary storage region for when the CPU **401** performs various kinds of processing. The ROM **403** stores an OS and other application software. In the present embodiment, the ROM **403** is a data-rewritable non-volatile memory typified by a flash memory. A communication unit **404** is an interface to which a LAN cable is connected, and data communication with each of the application server **100**, the management server **200**, and the management server **300** is performed via a router (not illustrated) or the network **1000**. The data communication may be performed wirelessly, for example, by an interface that supports wireless communication.
(25) A display unit **408** is configured by, for example, an LCD, and displays various kinds of information. An operation unit **409** includes a button, a touch panel, and the like for the user to perform various kinds of input operations. A printing apparatus **406** processes job data received by the communication unit **404** and performs printing on a document sheet. The storage device **430** stores job data received by the communication unit **404**. A system bus **407** exchanges data between the CPU **401**, the RAM **402**, the ROM **403**, the communication unit **404**, the printing apparatus **406**, the display unit **408**, the operation unit **409**, and the like. The processing of the communication module **410** and the print module **420** of the printer **400** is realized by the CPU **401** reading and executing a program necessary for the processing.
(26) Connection of Printer not Having Server Switching Function and Management Server (FIG. **4**)
(27) Next, processing in which in the service of the configuration of FIG. **1**, a printer **400′** not having the server switching function is connected to a management server and registered as a cloud printing printer will be described. An address (includes a URL or the like) of a management server **200′** is registered in the printer **400′** registered in the management server **200′** as a communication destination for when performing cloud printing. The printer **400′** obtains a job list and job data from the management server **200′** and performs cloud printing. An apostrophe (′) is attached to the reference numeral of the printer and the management server in the present description, because functions of the printer and the management server of FIG. **4** and the printer and the management servers of FIGS. **5** to **8** are different. In this example, the printer does not have the server switching function, and the management server also does not support the server switching function of the printer. Thus, an apostrophe (′) following a reference numeral indicates that the server switching function is not supported.
(28) FIG. **4** is a sequence diagram illustrating processing for registering the printer **400′**, which does not have the server switching function, in the management server **200′**. A printer not having the server switching function connects only to a particular management server. In this example, assume that the management server **200′** is a management server to which a printer not having the server switching function connects. Here, the management server **200′** by which the printer **400′** is managed is predetermined, and for example, an address, such as a URL of the management server **200′**, is registered as an access destination in the ROM **403**.
(29) The user performs a server connection operation with respect to the printer **400′** (step or operation S**601**). The communication module **410** of the printer **400′**, after it has received the connection operation, transmits a printer registration request to the management server **200′**, which is registered in advance as a transmission destination (step S**602**). In step S**602**, identification information of the printer **400′** and, if necessary, password information and the like, for example, are transmitted together with the printer registration request.
(30) The management server **200′**, after receiving the registration request, generates information of the requesting printer **400′** and authentication information, which is also referred to as a token or the like for the printer **400′** to connect to the management server **200′**. The generated information is associated with the requesting printer **400′** and stored in the database **210** (step S**603**). The management server **200′** transmits the generated authentication information to the printer **400′** as a response to the registration request (step S**604**). Upon receiving the response transmitted in step S**604**, the printer **400′** displays server registration completion to the user on the display unit **408**

and notifies the user that registration is complete (step S**605**). The received authentication information is stored to be used when accessing the management server **200**′.
(31) The management server **200**′ notifies the application server **100** and the printer **400**′ that printer registration is complete (step S**606**). This makes it possible for the application server **100** to recognize that the printer **400**′ is registered to the management server **200**′ and is being managed. From step S**606** onward, the printer **400**′ transmits information of various kinds of events (such as an error, an occurrence of an alert, and a change in the remaining amount of ink) that have occurred in the printer **400**′ to the management server **200**′ with the authentication information received in step S**604** attached, each time an event occurs (step S**607**). The printer **400**′ is also capable of obtaining a job list and job data from the management server **200**′ and performing printing.
(32) Upon receiving transmission of various kinds of event information or a request from the printer **400**′, the management server **200**′ confirms authentication information attached to the information or the request and processes the request as information of the printer associated with the authentication information (step S**608**). In this example, regarding the server connection establishment, the printer **400**′ notifies the user upon receiving the authentication information from the management server **200**′; however, the printer **400**′ may notify the user immediately after transmitting the registration request in step S**602** without waiting for step S**604**. That is, the user may be notified of the connection establishment asynchronously to registration processing with the management server **200**′.
(33) By the printer **400**′ being registered in the preset management server **200**′ with the above procedure, a cloud printing service in which the printer **400**′ is used can be provided.
(34) Initial Registration of Printer Having Server Switching Function in Management Server (FIG. **5**)
(35) Next, processing in which in the cloud printing service of the configuration of FIG. **1**, the printer **400**, which has the server switching function, is connected to and registered in the management server **300** determined as a connection destination of the printer **400** from among a plurality of management servers will be described with reference to FIG. **5**. In this example, the printer **400** has the server switching function from when it is manufactured, and the management servers **200** and **300** perform operations corresponding to the printer **400** having the switching function. Whether the printer **400** has the server switching function from when it is manufactured is indicated by a firmware update flag (also called a ROM update flag), which the printer has and, if necessary, transmits with a request. In the printer **400** provided with the server switching function by a firmware update, the firmware update flag is, for example, set to on, and in the printer **400** provided with the server switching function at the time of manufacturing, the firmware update flag is set to off. In addition, the operation of the printer **400** is performed by the CPU **401**, and the operation of each server is performed by the CPU **301**. This is similar for FIGS. **6** to **8**. In the following description, it is assumed that the firmware update flag is off. In this case, it is assumed that the printer **400** does not include a firmware update flag in a request to the server.
(36) As will be described later, when the firmware update flag is set to on, it can be said that the server switching function is enabled, and so, the firmware update flag may be referred to as a server switching flag or server switching information. The firmware update flag may be referred to as hardware update information. The flag indicates that firmware of the printer has been updated and thus can also be referred to as a firmware update event or a ROM update event. Alternatively, the flag can be referred to as a server switching event.
(37) The user performs a server connection operation by operating the operation unit **409** of the printer **400** (step S**701**). The communication module **410** of the printer **400**, after having received the connection operation, transmits a connection destination query to the management server **200** (step S**702**). The management server **200** is registered in advance, for example, in the ROM **403** as an access destination for a query.
(38) The control application **220** of the management server **200**, after having received the query,

determines a management server of the printer **400** based on printer destination information included in query information and an installation position of the printer **400** inferred from an IP address or the like of a query transmission source. The management server and the installation position of the printer may be associated in advance. In this example, a determined management server is the management server **300**. The control application **220** returns a connection destination of the determined management server, such as a URL 1 (step S**703**). The URL 1 may be a connection destination for registering the printer **400** in the management server **300**. FIG. **5** illustrates a case where the connection destination determined and returned in step S**703** is the management server **300**.
(39) The communication module **410** of the printer **400**, after having received the connection destination URL 1, transmits a printer registration request to the connection destination URL 1 (step S**704**). In this example, the URL 1 is the management server **300**, and so, the printer registration request is transmitted to the management server **300**. The identification information of the printer **400** is transmitted with the printer registration request.
(40) The management server **300**, after having received the registration request, transmits to the global database **600** a request for obtaining the printer information of the requesting printer **400** and the management server information of the printer **400** (step S**705**). The request is also referred to as a printer management information request. The printer information includes information related to the printer **400**, such as identification information. The management server information includes information, such as an identifier indicating the management server of the printer **400**. Information including the printer information and the management server information is referred to as printer management information. In this example, the printer **400** is not registered. Therefore, the global database **600** returns a response indicating that the printer is not registered (step S**706**). The response is also referred to as a printer management information request. Then, the management server **300**, after having received in step S**706** the response indicating that the printer **400** is not registered in the global database **600**, determines whether a firmware update flag is included in the registration request of step S**704**. In this case, the firmware update flag is not included. In such a case, the management server **300** registers the printer management information of the printer **400** in the global database **600**. FIG. **5** illustrates obtainment and registration of the printer management information collectively in steps S**705** and S**706**.
(41) The management server **300** also generates connection destination information, such as a URL 2, for the printer **400** to access the management server **300** and authentication information, such as a token, for the printer **400** to connect to the management server **300**. The generated information is associated with the printer **400** and registered in the database **310** (step S**707**). The control application **320** of the management server **300** transmits the generated connection destination information, such as the connection destination URL 2, and authentication information as a response to the printer **400** (step S**708**). The URL 2 may be a connection destination for accessing the management server **300** when the printer **400** provides the cloud printing service. Further, in FIG. **5**, both the URL 1 and the URL 2 are the management server **300** itself; however, depending on registration information of the application server **100**, they may be a management server (not illustrated) other than the management server **300** or the management server **200**.
(42) Upon receiving the connection destination URL 2 and the authentication information, the printer **400** stores the information and notifies the user by displaying on the display unit **408** that connection is established (step S**709**). The management server **300** notifies the application server **100** that the registration of the printer **400** is complete (step S**710**).
(43) This makes it possible for the application server **100** to recognize with which of a plurality of management servers the printer **400** is connected. Thereafter, the printer **400** transmits information of various kinds of events (such as an error, an occurrence of an alert, and a change in the remaining amount of ink) that have occurred in the printer **400** to the management server **300** with the authentication information received in step S**708** attached (step S**711**). When the management

server **300** receives transmission of various kinds of event information or a request from the printer **400**, the management server **300** confirms the attached authentication information and processes the request as information of the printer corresponding to the authentication information (step S**712**).

(44) A client can obtain, from the management server **300**, information indicating that the printer **400** is included as a printer for the cloud printing service, specify the printer **400**, and instruct cloud printing. The printer **400** can obtain, from the management server **300**, job data to be printed on the printer **400** and print the job data.

(45) In the above-described example, a method in which the printer **400** queries the management server **200** in the determination of the connection destination of the printer **400** is described; however, a configuration may be such that the management server **300** is queried. Alternatively, a configuration may be such that, for example, a server dedicated to processing only connection destination queries is installed and that server is queried.

(46) With such a procedure, a printer having the server switching function can be registered in a management server determined for that printer. Once registered, the printer can perform cloud printing by communicating with that management server.

(47) Re-Obtainment of Management Server Information by Registered Printer Having Server Switching Function (FIG. **6**)

(48) Next, a flow in which in the service of the configuration of FIG. **1**, the connection destination URLs 1 and 2 and the authentication information are reobtained when the printer **400** and the management server **300**, which is a management server of that printer, are already registered in the global database **600** will be described with reference to FIG. **6**. In this example, the printer **400** has the server switching function from when it is manufactured, and the management servers **200** and **300** perform operations corresponding to the printer **400** having the switching function. Whether the printer **400** has the server switching function from when it is manufactured is indicated by a firmware update flag, which is transmitted, as necessary, with a request. In the following description, it is assumed that the firmware update flag is off. In this case, it is assumed that the printer **400** does not include a firmware update flag in a request to the server. In addition, the operation of the printer **400** is performed by the CPU **401**, and the operation of each server is performed by the CPU **301**.

(49) If a URL (connection destination information) and authentication information of a management server of the printer **400** are stored in a volatile memory, the stored information may be lost, for example, when the power of the printer **400** is turned off and then turned on again. Alternatively, information stored in a hard disk and the like may be lost due to damage. A procedure of FIG. **6** is performed, for example, when a URL and authentication information of a connection destination management server determined by a management server with the procedure of FIG. **5** are lost from the printer **400**.

(50) Even when at least either a URL or authentication information of the connection destination management server in the printer **400** is lost, the printer **400** communicates with the management server **200** and the management server **300** with an interface similar to that described in FIG. **5**.

(51) When connection destination information or authentication information is lost or damaged, the printer **400** determines whether a connection destination needs to be queried again (step S**801**). It is determined whether information is lost or damaged when connection destination information and authentication information are referenced, such as when a self-diagnostic program is periodically run and when the user attempts to obtain a job with the printer **400**.

(52) The communication module **410** of the printer **400** transmits a connection destination server query to the management server **200** (step S**802**). Address information of the management server **200** may be protected from loss or damage by being stored in a non-volatile medium, such as a flash memory or a ROM.

(53) The control application **220** of the management server **200**, after having received the query,

determines a management server of the printer **400** based on printer destination information included in query information and an installation position of the printer **400** inferred from an Internet Protocol (IP) address or the like of a query transmission source. The management server and the installation position of the printer may be associated in advance. The control application **220** returns a connection destination of the determined management server, such as the URL 1 (step S**803**). The URL 1 may be a connection destination for registering the printer **400** in the management server **300**. Similarly to FIG. **5**, FIG. **6** illustrates a case where the connection destination determined and returned is the management server **300**.

(54) The communication module **410** of the printer **400**, after having received the connection destination URL 1, transmits a printer registration request to the URL 1, that is, the management server **300** (step S**804**).

(55) The management server **300**, after having received the registration request, transmits to the global database **600** a printer management information request for obtaining the printer information of the requesting printer **400** and the management server information of the printer **400** (step S**805**). In this case, the printer **400** is already registered, and so, the global database **600** returns registered printer information and management server information as a printer management information response (step S**806**). The management server **300**, after having received the response, obtains connection destination information of the printer **400** and authentication information from the database **310** (step S**807**). The obtained management server information includes the authentication information and the connection destination URL 2. The management server **300** transmits the authentication information and the connection destination URL 2 obtained from the database **310** to the printer **400** as a response to the registration request (step S**808**).

(56) Thereafter, the printer **400** can once again transmit information of various kinds of events (such as an error, an occurrence of an alert, and a change in the remaining amount of ink) that have occurred in the printer **400** to the management server **300** with the authentication information received in step S**808** attached (step S**809**).

(57) In the above example, a method in which the printer **400** issues a registration request triggered by loss or damage of the connection destination URL and the authentication information, and then reobtains the management server information in response to the registration request has been described. However, a configuration may be such that information is transmitted without detection of loss or damage, and when a result indicates failure, a connection destination is queried, triggered by the failure. Alternatively, a configuration may be such that before an attempt is made to transmit information, it is confirmed that authentication information is present in the printer **400**. Alternatively, a configuration may be such that whenever the printer **400** is in a particular state, the printer **400** performs a connection destination query of step S**802** with respect to the management server **200**, a dedicated server of that type, or the like as part of the initialization processing. By performing such processing described in FIG. **6**, even at the time of reobtainment, a connection destination URL and authentication information necessary for connection can be passed to the printer **400** with an interface similar to that described in FIG. **5**.

(58) Registration in Management Server by Printer to which Server Switching Function has been Added (FIG. **7**)

(59) Next, processing by which the printer **400**, after having obtained the server switching function by a firmware update (also referred to as a ROM update), connects with the management server **300** and makes a registration request will be described with reference to FIG. **7**. This procedure is also realized in the cloud printing service of the configuration of FIG. **1**. In this procedure, the printer **400** does not have the server switching function at the time of manufacturing, and the server switching function is added by the firmware update. The printer **400** is not registered in the cloud printing system until the firmware update. FIG. **7** illustrates a procedure for registration in such a case. In the drawing and in the following text, firmware is denoted as FW. When querying a connection destination management server, the printer **400** informs the management server **300** that

it is a printer that has additionally obtained the server switching function by using a firmware updated flag. The printer **400** is not registered, and so there is no printer information in the management server **200**, and similarly to a printer having a function for switching connection destinations from when it is manufactured described in FIG. **5**, the printer **400** is registered by connecting to the management server **300**.

(60) The user performs an FW update operation with respect to the printer **400** (step S**901**). The printer **400** performs a firmware update in response to the firmware update operation (step S**902**). At this time, pre-update firmware does not have a function for switching registered management servers, and that function is added to post-update firmware. The printer **400** notifies a user of firmware update completion by, for example, displaying firmware update completion to the user using the display unit **408** (step S**903**).

(61) After the firmware update is complete, the user performs a server connection operation with respect to the printer **400** (step S**904**). The communication module **410** of the printer **400** queries the management server **200** for a connection destination (step S**905**). The management server **200**, which is a destination of the connection destination query, is predetermined, and that destination is stored in the printer **400**. This destination may be stored, for example, with the firmware update or before the firmware update.

(62) The control application **220** of the management server **200**, after having received the query, determines a management server of the printer **400** based on printer destination information included in query information and an installation position of the printer **400** inferred from an IP address or the like of a query request transmission source. The connection destination URL 1 of the determined management server is returned (step S**906**). FIG. **7** illustrates a case where the determined management server is the management server **300**. The communication module **410** of the printer **400**, after having received the connection destination URL 1, transmits a printer registration request and a firmware update flag to the management server **300** (step S**907**).

(63) The management server **300**, after having received the registration request, transmits to the global database **600** a printer management information request for obtaining the printer information and the management server information of the requesting printer **400** (step S**908**). This request also includes information related to the printer **400**, such as identification information. In this example, the printer **400** is not registered. If it is determined that the printer **400** is not registered, the global database **600** returns a printer management information response indicating that the printer **400** is not registered (step S**909**). The management server **300**, after having received that response, determines whether a firmware update flag is included in the registration request (step S**910**). If the management server **300** determines that a firmware update flag is included, the management server **300** transmits a request for confirming (or obtaining) whether the connection destination information of the printer **400** and the authentication information (collectively referred to as the connection information) are held to the management server **200** (step S**911**). This is because the printer **400** may have been connected to the management server **200** prior to the firmware update. The management server to which the printer **400** may have been connected prior to the firmware update can be identified based on, for example, the shipping destination of the printer **400**.

(64) The management server **200**, after having received the request, determines whether the connection information of the printer **400** is stored in the management server **200** (step S**912**). In this example, corresponding information is not registered. In such a case, the management server **200** registers in the global database **600** the management server information indicating that the management server **300** is the management server of the printer **400** (step S**913**). The management server of the printer **400** is determined by the management server **200**, and so, in step S**913**, the printer information and the management server information of the printer **400** may be registered based on that determination. In addition, the management server **200** returns to the management server **300** a determination result (e.g., there is no corresponding registration information in this case) in the management server **200** (step S**914**). The management server **300** returns to the printer

400 the authentication information and the connection destination URL 2 of the management server **300** registered as the management server of the printer **400** (step S**915**).

(65) Upon receiving the connection destination URL 2 and the authentication information of the connection destination URL 2 from the management server **300**, the printer **400** notifies a user that the server connection is established by displaying it on the display unit **408** to the user (step S**916**).

(66) In this example, the printer **400** in which firmware has been updated can thus be newly registered to the cloud printing system. Then, the management server determined as the management server of the printer **400** is registered in the global database, and the connection information is registered in the determined management server **300**. The connection information is also stored in the printer **400** and is referenced for the cloud printing service in which the printer **400** is used.

(67) Switching of Management Servers by Printer to which Server Switching Function has been Added (FIG. **8**)

(68) In the procedure of FIG. **4**, the printer **400** is registered to the management server **200** until the server switching function is obtained by a firmware update and performs printing with the cloud printing service under the management of the management server **200**. That is, event information is transmitted to the management server **200**, and print job data is obtained from the management server **200** and printed. FIG. **8** is a diagram of a sequence from the server switching function being added to the printer **400** registered in the management server **200** according to the procedure of FIG. **4** by a firmware update until a transmission destination of event information being switched over to the management server **300**, which is a new connection destination. That is, it illustrates a method of switching servers. In this example, a management server determined from the shipping destination or the like of the printer **400** is changed from the management server **200** to the management server **300**. However, since the printer **400** has been registered to the management server **200**, even if the new management server **300** is installed, the printer **400** is neither managed by nor communicates with the management server **300**. In an example of FIG. **8**, even in such a case, the new management server **300** can be registered as the management server of the cloud printing service. When querying a connection destination management server, the printer **400** informs the management server side that it is a printer that has obtained the server switching function by a firmware update by using a firmware update flag. Thus, the connection information of the printer **400** stored in the management server **200** is transferred to the management server **300**.

(69) The printer **400** transmits event information to the management server **200** using the connection destination URL and the authentication information obtained by the processing described in FIG. **4** (step S**1001**). The user performs a firmware update operation with respect to the printer **400** (step S**1002**). The printer **400** performs a firmware update in response to the firmware update operation (step S**1003**). The printer **400** informs the user that the ROM update is complete by displaying so on the display unit **408** (step S**1004**). Even after step S**1004**, the printer **400** transmits event information to the management server **200** according to the stored connection information (step S**1005**). In addition, the printer **400** obtains job data from the management server **200** and prints the job data.

(70) After step S**1005**, the connection information including the connection destination URL and the authentication information present in the printer **400** is cleared (step S**1006**). The user may intentionally cause the information to be cleared. For example, a configuration may be such that a user interface for server switching is provided by the updated firmware, and the connection destination URL and the authentication information are cleared as a result of operation on the user interface. If clearing is possible by a power-off, a power-off/on operation may be performed. The printer **400** queries the management server **200** for a connection destination, triggered by the information being cleared (step S**1007**). The management server **200**, which is a destination of the connection destination query, is predetermined, and that destination is stored in the printer **400**.

This destination may be stored, for example, with the firmware update or from before the firmware update.

(71) The control application **220** of the management server **200**, after having received the query, determines a management server of the printer **400** based on printer destination information included in query information and an installation position of the printer **400** inferred from an IP address or the like of a query request transmission source. The connection destination URL 1 of the determined management server is returned (step S**1008**). FIG. **8** illustrates a case where the determined connection destination is the management server **300**. The communication module **410** of the printer **400**, after having received the connection destination URL 1, transmits a printer registration request and a firmware update flag to the management server **300** (step S**1009**).

(72) The control application **320** in the management server **300**, after having received the registration request, transmits to the global database **600** a request for obtaining the printer information and the management server information of the requesting printer **400** (step S**1010**). This request also includes information related to the printer **400**, such as identification information. As is evident from when FIG. **4** is referenced, in the procedure of FIG. **4**, the global database **600** does not appear, and so, the printer **400** is not registered. If it is determined that the printer **400** is not registered, the global database **600** returns a response indicating that the printer **400** is not registered (step S**1011**).

(73) The control application **320** of the management server **300**, after having received that response, determines whether the registration request transmitted in step S**1009** includes a firmware update flag (step S**1012**). When the control application **320** of the management server **300** determines that there is a firmware update flag, the control application **320** transmits to the management server **200** a request for confirming (or obtaining) whether the connection information of the printer **400** prior to the firmware update is stored (step S**1013**). This is because the printer **400** may have been connected to the management server **200** prior to the firmware update. The management server to which the printer **400** may have been connected prior to the firmware update can be identified based on, for example, the shipping destination of the printer **400**.

(74) The control application **220** of the management server **200**, after having received the request, determines whether the connection information of the printer **400** is stored in the database **210** (step S**1014**). In this example, corresponding information is stored. When the control application **220** of the management server **200** determines that the connection information of the printer **400** is stored, the control application **220** writes information (transfer-in-progress information) indicating that the global database **600** is being transferred (step S**1016**). When the control application **220** of the management server **200** writes in step S**1016** that transfer is in progress, the control application **220** communicates with the control application **320** in the management server **300** and performs (step S**1017**) processing for transferring information related to the printer **400** asynchronously to the flow that is indicated by a solid line in FIG. **8**. The asynchronously-performed processing is illustrated by a dotted line in the drawing. The transfer processing includes, for example, duplication of information from the management server **200** to the management server **300**. Information to be duplicated may include, for example, a job list and job data of print jobs to be printed in the printer **400** and their print setting information. The control application **220** of the management server **200** returns the connection information of the printer **400** stored in the database **210** of the management server **200** to the control application **320** of the management server **300** (step S**1018**).

(75) When the control application **320** of the management server **300** receives the response of step S**1018**, the control application **320** returns (step S**1019**) to the printer **400** the received connection information (authentication information and connection destination URL 2) of the printer **400**. This connection information includes the authentication information and the connection destination URL 2 for the management server **200**. Furthermore, when the transfer processing has proceeded to a predetermined step or operation, the management server **300** updates a “transfer-in-progress”

state of the information related to the printer, which has been stored in the global database **600** in step S**1016**, to a "transfer-ready" state (step S**1020**). With transfer ready being written, the printer management information may be registered such that the management server that manages the printer **400** is the new management server **300**.

(76) Here, the predetermined step refers to a step or an operation in which even if a request is made to the management server **300** which is a transfer destination of the printer **400**, a difference between the information stored in the management server **200** and the management server **300** will not result in abnormal processing. For example, if servers are switched before the start of duplication of unprocessed data (e.g., job data that the printer **400** has yet to obtain), an attempt may be made to obtain that unprocessed data from a post-switch server. In this case, the data to be obtained is not on the post-switch server, and the request for obtaining data results in an error. To prevent these kinds of situations, a transition to the transfer-ready state may be made, for example, after the start of duplication of unprocessed data (e.g., unobtained job data and print setting information) from the management server **200** to the management server **300**. This predetermined step can be said to be a timing at which the transfer has become ready. It can be recognized that the transfer is ready not only by the management server **300** but also by the management server **200**. Therefore, step S**1020** may be performed by the management server **200**.

(77) When the transfer is ready, after step S**1020**, the management server **200** transmits a management server transfer notification to the printer **400**. The authentication information and the connection destination URL 2 (i.e., the connection information) of the management server **300** are transmitted with the notification (step S**1021**). With this, it is notified that the management server of the printer **400** has been transferred from the management server **200** to the management server **300**. The connection information of the management server **300** may be obtained by the management server **200** making a request to the management server **300**. At this time, in response to the request, the management server **300** may generate and store the connection information. That is, the printer **400** may be registered at this time. In addition, the management server **200**, after having transmitted the connection information of the management server **300** to the printer **400**, may clear registered information of the printer **400**, such as the connection information.

(78) After the transfer is ready and the management server **200** transmits a transfer notification to the printer **400** in step S**1021**, the management server **300** transmits a transfer-ready notification to the application server **100** (step S**1022**). With this, it is notified that the management server of the printer **400** has been transferred from the management server **200** to the management server **300**. After step S**1022**, the printer **400** transmits event information and the like that have so far been transmitted to the management server **200** to the management server **300** (step S**1023**).

(79) Steps S**1020** and S**1022** are executed by the management server **300**, and step S**1021** is executed by the management server **200**. According to the above description, these steps are performed in order of step S**1020**, step S**1021**, and step S**1022**, and so, the management server **200**, which executes step S**1021**, needs to know that step S**1020** has been executed by the management server **300**. At this time, the management server **200** also needs to know the connection information of the management server **300**. The management server **300**, which executes step S**1022**, also needs to know that step S**1021** has been executed by the management server **200**. Therefore, although not clearly indicated in FIG. **8**, the management server **300**, after having executed step S**1020**, may notify the management server **200** that step S**1020** has been executed. Similarly, the management server **200**, after having executed step S**1021**, may notify the management server **300** that step S**1021** has been executed.

(80) Further, in step S**1022** of FIG. **8**, the management server **300** notifies the printer **400** of the management server transfer; however, a case where this notification fails due to network/protocol failure is conceivable. At that time, the management server **200** and the printer **400** communicates until a connection destination query of step S**1007** is received from the printer **400** again, and the post-transfer information is passed by transmission of the authentication information and the

connection destination URL 2 of the management server **200** in step S**1018**. Further, although the transmission source of steps S**1019** and S**1022** of FIG. **8** is the management server **300**, which is the transfer destination, the transmission may be performed from the management server **200**, which is the transfer source.
(81) In addition, in step S**1009**, the printer **400** transmits a firmware update flag to the management server **300**, and in step S**1012**, the firmware update flag is confirmed. In contrast to this, the firmware update flag need not be present. In such cases, a configuration may be taken so as to skip the processing of step S**1012** and always perform the processing of step S**1013**. At that time, the processing of step S**1013** needs to be performed even though step S**1013** would be unnecessary as in FIG. **5** if the printer has the server switching function at the time of manufacturing. However, there is an advantage that processing can be standardized so long as the printer has the server switching function, regardless of whether it is from the time of manufacturing or it has been obtained by a firmware update. Further, such a configuration makes it possible for the printer **400** having the server switching function from the time of manufacturing to transfer the management server without setting a firmware update flag.
(82) In addition, as described in the procedure of FIG. **5**, a printer that has been provided with the server switching function from the time of manufacturing is incapable of handling transferring of the management server as is. However, by setting a firmware update flag to on, the procedure described in FIG. **8** is adopted, and thus, the management servers can be switched.
(83) As described above, procedures for when management server transfer occurs have been described with respect to the cloud printing system as a whole. Here, a printer having the server switching function, a global database, and management servers will be described anew.
(84) Server Switching Function of Printer
(85) A printer having the server switching function transmits a connection query to a predetermined management server (in the above examples, the management server **200**) according to a server connection operation by the user. In response to this connection query, the predetermined server determines a management server of the querying printer and returns connection destination information (such as a URL), which is an address of the management server, to the printer. The printer, after having received the connection destination, transmits a registration request to the received connection destination. In this way, as compared to a printer that does not have the server switching function, the printer that has that function first executes processing for obtaining information of a registration request transmission destination from a specific server. This step makes it possible for the management server corresponding to the printer to not be fixed, and for the cloud printing service provider side to determine the server, as appropriate, and change the server.
(86) Global Database
(87) By a global database being provided, even when a plurality of management servers manage printers, for example, for each region, it is possible to manage printer registration statuses across the entirety of a plurality of management servers. The registration statuses allow centralized management of presence or absence of registration or the like as well as which printer is registered in which management server.
(88) Processing of Connection Request by Management Server **300**
(89) In the following, the processing of the management server **300** after it has received a registration request will be described again with reference to FIGS. **9** and **10**. The processing of these drawings is executed by the CPU **301** of the management server **300**.
(90) The management server **300**, after having received a registration request, performs the processing of FIG. **9**. First, an attempt to obtain printer management information from the global database **600** is made (step S**9001**). It is determined whether obtainment is successful (step S**9002**) and, if successful, it is determined whether a firmware update flag is included in the request (step S**9003**). When a firmware update flag is always included and is set to either “on” or “off”,

"included" includes a state in which the flag is set to on.

(91) If a firmware update flag is included, a request for confirming presence or absence of connection information (URL and authentication information of the management server) of the printer, which is the source of the registration request, is issued to the management server **200** (step S**9004**). This request may be a request for obtaining connection information. If there is corresponding connection information for the request, that information is returned; otherwise, something to that effect is returned. Then, presence or absence of connection information is determined by the response being tested (step S**9005**). If the response contains connection information, the received connection information is transmitted to the printer, which is the source of the registration request (step S**9006**). The procedure until step S**9006** is equivalent to the case of FIG. **8**.

(92) Meanwhile, if connection information is not included, connection information including an address (e.g. URL) and authentication information (e.g. token) of the management server is generated, stored, and managed by that management server (step S**9007**). Then, that connection information is transmitted to the printer, which is the source of the registration request (step S**9008**). The procedure from after step S**9005** until step S**9008** is equivalent to the case of FIG. **7**.

(93) If it is determined in step S**9003** that a firmware update flag is not included in the registration request, printer management information including information of the printer, which is the source of the registration request, and management server information of the management server to which the registration request has been made is generated. Then, that information is registered in the global database **600** (step S**9009**). Thereafter, the processing from step S**9007** onward is executed. The procedure branching from step S**9003** to step S**9009** is equivalent to the case of FIG. **5**.

(94) When it is determined in step S**9002** that printer management information has been successfully obtained, connection information that should be stored in the management server is obtained (step S**9010**) from that management server, the management server being that of the printer, which is the source of the registration request, and included in the printer management information. Then, in step S**9008**, that information is transmitted to the printer, which is the source of the registration request.

(95) FIG. **10** illustrates a procedure of the transfer processing that corresponds to steps S**1017** and S**1020** to S**1022** of FIG. **8** and which is executed asynchronously to the processing of FIG. **9**. The transfer processing is started with the management server **200** (step S**10001**). The transfer processing is started by the management server **200** after the determination result of step S**1014** and the processing of step S**1016** corresponding thereto. This transfer processing includes the processing for duplicating information related to the printer, which is the transmission source of the registration request, from a pre-switch management server to a post-switch management server. Assume that the transfer processing including this duplication continues to be performed until transferring of target data is complete.

(96) Until the transferring of data advances and reaches the predetermined step described in FIG. **8** is awaited (step S**10002**), and when the step is reached, transfer ready is written into the global database **600** (step S**1003**). The connection information of a post-switch management server is transmitted to the printer, which is the source of the registration request (step S**10004**). Although this step is performed by the management server **200** in FIG. **8**, the description thereof assumes that it may be performed by the management server **300**, and in FIG. **10**, it is assumed that this step is performed by the management server **300**. In this case, the connection information for connecting to the management server may be generated and transmitted in step S**10004**.

(97) Finally, a management server transfer notification is transmitted also to the application server **100** (step S**10005**). This notification may include, for example, printer information of the printer for which the management servers have been switched and information of the new management server.

Effects of Present Embodiment

(98) According to the above configurations and procedures, the cloud printing system can transfer data related to a printer between management servers necessary for management server switching without interrupting provision of the cloud printing service. Furthermore, if the above-described procedures are applied to a client apparatus that is not a printer, transfer of data for that client apparatus can similarly be performed while continuing a service.
(99) In addition, even if not all of the data on the printer has been transferred, so long as the transferring has advanced to a predetermined step, it is possible to transfer to a new management server, and thus, it is possible to swiftly advance management server transfer.
(100) Furthermore, so long as a printer has the server switching function, a management server, which is a switchover destination, can be determined and changed by a system administrator without the firmware of the printer being changed.
(101) In addition, the registration statuses of all of the printers registered in management servers can be managed in a centralized manner by a global database, and so it is possible to prevent omissions in management attributable to management server transfer.
(102) Furthermore, a printer and its management server are associated in a centralized manner by a particular server, and so, it becomes possible to achieve simplification of management work and increased efficiency.
OTHER EMBODIMENTS
(103) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(104) While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(105) This application claims the benefit of Japanese Patent Application No. 2022-109064, filed Jul. 6, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A network system including a device, a first server, and a second server, the device transmitting operation information to a connected server, the second server comprising: at least one second processor, wherein at least one second program executed by the at least one second processor causes the second server to: transmit, to the first server, a request for first connection information for the device to connect to the first server based on reception of a registration request

accompanying a firmware update from the device, and the first server comprising: at least one first processor, wherein at least one first program executed by the at least one first processor causes the first server to: after a reception of the request for the first connection information, start transferring, to the second server, data related to the device, and transmit to the second server the first connection information in response to the request for the first connection information, wherein the at least one second program causes the second server to further transmit to the device the first connection information received from the first server, and wherein when the transferring of data unobtained by the device included in the data related to the device is started, second connection information for connecting to the second server is transmitted from either the first server or the second server to the device.

2. The network system according to claim 1, wherein the device queries a predetermined server for a connection destination and, in a case where the second server is returned as the connection destination, transmits the registration request to the second server.

3. The network system according to claim 1, wherein in a case where the first connection information that had been stored is lost, the device transmits the registration request.

4. The network system according to claim 1, wherein the data related to the device and to be transferred includes data to be obtained by the device.

5. The network system according to claim 1, wherein the registration request is received from the device when the firmware update of the device has occurred.

6. The network system according to claim 1, wherein the first connection information includes authentication information for connecting to the first server, and the second connection information includes authentication information for connecting to the second server.

7. The network system according to claim 6, wherein the first connection information includes an address of the first server, and the second connection information includes an address of the second server.

8. The network system according to claim 1, wherein the network system further includes an application server configured to provide data to the first server or the second server, and after the second connection information is transmitted to the device, the second server further notifies the application server that the device is registered in the second server.

9. The network system according to claim 1, wherein when the transferring of the data unobtained by the device included in the data related to the device is started, the first server transmits to the device the second connection information received from the second server.

10. The network system according to claim 1, wherein when the transferring of the data unobtained by the device included in the data related to the device is started, the second server generates the second connection information and transmits the second connection information to the device.

11. A method of switching servers in a network system including a device, a first server, and a second server, the method comprising: the device transmitting operation information to a connected server, the second server transmitting, to the first server, a request for obtaining first connection information for the device to connect to the first server based on reception of a registration request accompanying a firmware update from the device, and the first server after a reception of the request for the first connection information, starting transferring of data related to the device to the second server, and transmitting to the second server the first connection information in response to the request for the first connection information, wherein the second server transmits to the device the first connection information received from the first server, and wherein when the transferring of data unobtained by the device included in the data related to the device is started, second connection information for connecting to the second server is transmitted from either the first server or the second server to the device.